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The Complexity Calculation for Group Decision Making Using TOPSIS Algorithm

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Abstract. The complexity calculation is one of the things to be considered in choosing an algorithm to be applied to a decision support system (DSS). This study analyzes the complexity of Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) algorithm. This algorithm is applied to the ranking process to resolve the problems associated with Multi-Criteria Decision Making (MCDM). MCDM is used for Group Decision Making (GDM) problems in making a decision. The end output of this study is making DM ranking resulted from decision-making. This problem also produces TOPSIS algorithm complexity used in group decision-making. This algorithm complexity is in the part of attribute normalization and weighting which results in $O(n^2)$. Whereas the complexity of a positive-negative ideal solution and V distance to A1, A2, A3, and A4 is $O(n)$ and the algorithm complexity ranking results in $O(1)$.

INTRODUCTION

Complexity is a part of computer sciences related to the usage of an algorithm in a particular application, such as group decision-making in this paper. The algorithm complexity is divided into two, i.e. the time complexity and space complexity. Time complexity is measured by the number of computing stages required to execute an algorithm as a function of a number of data n (input size). Whereas the space complexity is measured by the memory used by the data structure contained in the algorithm as a function of a number of data n [1].

Algorithm complexity is a step-by-step procedure for the analysis. Algorithm is used for automatic data analysis, processing, and reasoning. Algorithm can be classified based on the amount of time required to complete compared to the size of the input. There is a range of varieties, e.g. Some algorithms complete in a linear time, relative to the size of input; others complete in the amount of time that is exponential or worse; and while others stop. The complexity analysis can choose an algorithm to be applied to the decision support system (DSS) [2]. Therefore, complexity is required to support the decision-making using a Multi-Criteria Decision Making (MCDM) [3–5]. One of the steps were taken in decision making is the formation of parameter-criteria, the completion of the decision model for weighting and alternative decision output. MCDM model can be used to solve decision-making problems using TOPSIS algorithm [3–7]. TOPSIS algorithm is very reliable in the computing process. TOPSIS is also very fast, easy to use and appropriate for decision-making [8,9] and for determining the most preferred option among all other options [10]. TOPSIS can be developed, and it can perform inversion for group decision-making [11,12]. Thus, TOPSIS algorithm is very appropriate for Multiple Attribute Group Decision-Making (MAGDM) approaches [13], which can compensate the accuracy, complexity and interpretability [9,14]. This paper is presented using four (4) parameters and 4 (four) alternatives by involving three (3) group decision making (GDM). Each group DM has the same data input criteria and different weights, thus, the decision ranking of each group DM

can be identified to be used in each DM ranking.

This paper also performed several testing to DM, including the following: normalizing matrix, searching for positive ideal solutions (PIS) A^+ and negative ideal solutions (NIS) A^- . These are to determine the normalized weight rating, finding the alternative distance A_i with PIS and NIS, as well as finding the proximity of each alternative output (V_i). Then the final stage sought the complexity by using TOPSIS algorithm to determine the speed of the process and stages of the use of this algorithm.

THEORETICAL BASIC

Complexity

A good algorithm is the one that is not necessarily true, but must be efficient. The algorithm efficiency is measured from the time for executing algorithms and the memory space requirements. An efficient algorithm is the one that minimizes the need for time and space. The need for time and space of an algorithm depends on the size of an input (n), which states the amount of processed data. The algorithm efficiency can be used to assess algorithms determining the best algorithm to solve the problems. The size used to describe the abstract model of this time or space measurement is algorithm complexity [15]. There are two kinds of algorithm complexities, i.e. time complexity and space complexity. Time complexity is $T(n)$, while space complexity is $S(n)$, whereas the size of data input for an algorithm is n . For example, in the array elements sorting algorithms, n is the number of array elements, whereas in matrix multiplication algorithm, n is the size of $n \times n$ matrices. First, in the space complexity, $S(n)$ is measured from the memory used by the data structure contained in the algorithm as a function of n input size. And second, in the time complexity, $T(n)$ is calculated based on the number of operation computing stages performed in an algorithm. In practice, the operation being analyzed is only typical operations that underlie an algorithm. For example, typical operating within the search algorithm in an array element comparison. Typical operation of sorting algorithm is the comparison and exchange of elements [16]. The algorithm complexity can also predict the future decision-making [17] and measure the automatic systems to high levels [18].

Technique for Order Preference by Similarity to Ideal Solution

TOPSIS method was first introduced by Yoon and Hwang in 1981. In general, TOPSIS procedure follows steps to make normalized decision matrix, weight normalization, positive ideal solutions (PIS) and negative ideal solutions (NIS). The distance between the alternative values are positive and negative matrices, and it determines the preference value for each alternative decision. This TOPSIS method requires A_i alternative in each normalized C_i criteria [8,10,19]. TOPSIS is based on a concept that each better alternative has shortest distance from positive ideal solution (PIS) and the one having the largest distance is the negative ideal solution (NIS) [19]. PIS presents the best solution which maximizes the benefit attribute and minimalizes the cost attribute whereas NIS presents the converse, i.e. negative solution minimizing benefits attribute and maximizing cost attribute [9]. Normalization is done as in Table 1 using equation (1).

TABLE 1. Alternative Suitability Rating of Each Parameter

Alternative	P1	P2	...	Pn
A1	A1 ₁	A2 ₁	...	A1 _n
A2	A2 ₁	A2 ₂	...	A2 _n
An	An ₁	An ₂	...	An _n

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}; \text{ if } i=1,2,\dots,m, \text{ and } j=1,2,\dots,n. \quad (1)$$

For each positive ideal solution (PIS) A^+ and negative ideal solution (NIS) A^- , normalized wight rating (y_{ij}) can be determined using equation (2), (3) dan (4).

$$(y_{ij}) = w_i r_{ij}; \text{ if } i = 1, 2, \dots, m, \text{ and } j = 1, 2, \dots, n \quad (2)$$

$$A^+ (\text{PIS}) = (y_1^+, y_2^+, \dots, y_n^+) \quad (3)$$

$$A^- (\text{NIS}) = (y_1^-, y_2^-, \dots, y_n^-) \quad (4)$$

where Y_j value results using equation (5) or (6).

$$Y_j^+ = \begin{cases} \max_i y_{ij} ; \text{ if } j \text{ for benefit attribute} \\ \min_i y_{ij} ; \text{ if } j \text{ for cost attribute} \end{cases} \quad (5)$$

$$Y_j^- = \begin{cases} \max_i y_{ij} ; \text{ if } j \text{ for benefit attribute} \\ \min_i y_{ij} ; \text{ if } j \text{ for cost attribute} \end{cases} \quad (6)$$

Then y_1^+ is conducted, $\max_1 y_{ij}$ value is + for benefit attribute and $\min_1 y_{ij}$ for cost attribute. And if $\min_1 y_{ij}$ value is for benefit attribute and $\max_1 y_{ij}$ for cost attribute. For alternative A_i distance with positive ideal solutions (PIS), it is calculated using equation (7).

$$S_i^+ = \sqrt{\sum_{j=1}^n (y_i - y_{ij}^+)^2} ; \quad i = 1, 2, \dots, m. \quad (7)$$

Then, the distance between A_i alternative with negative ideal solutions (NIS) is calculated using equation (8).

$$S_i^- = \sqrt{\sum_{j=1}^n (y_i - y_{ij}^-)^2} ; \quad i = 1, 2, \dots, m. \quad (8)$$

The preference value for each alternative (V_i) is calculated using equation (9).

$$V_i = \frac{S_i^-}{S_i^- + S_i^+} ; \quad i = 1, 2, \dots, m. \quad (9)$$

TOPSIS Algorithm Complexity

This section will calculate the time complexity of TOPSIS algorithm. The complexity analysis is carried out at several stages of TOPSIS algorithm. The complexity program and analysis of TOPSIS algorithm are as in Fig. 1, 2, 3, and 4.

<pre> 1 R = zeros (n,n); 2 Y = zeros (n,n); 3 for j=1:n; 4 TotX = 0; 5 for i=1:n; 6 Totx = Totx + X(i,j)*X(i,j); 7 end; 8 LX(j) = sqrt(Totx); 9 R(:,j) = X(:,j) ./ LX(j); 10 Y(:,j) = W(j)*R(:,j); 11 end;</pre>	• Line 1 $\rightarrow O(1)$ • Line 2 $\rightarrow O(1)$ • Line 6 $\rightarrow O(1)$ • Line 4 $\rightarrow n. O(1) = O(n). O(1) = O(n.1) = O(n)$ • Line 5 $\rightarrow n. O(1) = O(n). O(1) = O(n.1) = O(n)$ • Line 7 $\rightarrow n. O(1) = O(n). O(1) = O(n.1) = O(n)$ • Line 8 $\rightarrow n. O(1) = O(n). O(1) = O(n.1) = O(n)$ • Line 9 $\rightarrow n. O(1) = O(n). O(1) = O(n.1) = O(n)$ • Line 10 $\rightarrow n. O(1) = O(n). O(1) = O(n.1) = O(n)$ • Line 3 $\rightarrow n. O(n) = O(n). O(n) = O(n.n) = O(n^2)$ • Line 11 $\rightarrow O(1)$
(a)	(b)

FIGURE 1. Algorithm complexity for stage 1 (a) attribute normalization and weighing (b) the complexity analysis.

Then, the result of algorithm complexity in stage 1 is $O(n^2)$.

<pre> 1 for j=1:n, 2 if K(j)==1, 3 PIS(j) = max (Y(:,j)); 4 NIS(j) = min (Y(:,j)); 5 else 6 NIS(j) = min (Y(:,j)); 7 PIS(j) = max (Y(:,j)); 8 end; 9 end;</pre>	• Line 3 to 7 $\rightarrow O(1)$ • Line 2 and 8 $\rightarrow O(1)$ • Line 1 $\rightarrow n. O(1) = O(n)$ • Line 9 $\rightarrow O(1)$
(a)	(b)

FIGURE 2. Algorithm complexity for stage 2 (a) finding PIS and NIS (b) the complexity analysis.

The result of algorithm complexity in stage 2 to find PIS and NIS is $O(n)$

<pre> 1 for i=1:n, 2 S1(i) = sqrt(sum((Y(I,:) - A1(1,:)).^2)); 3 S2(i) = sqrt(sum((Y(I,:) - A2(1,:)).^2)); 4 S3(i) = sqrt(sum((Y(I,:) - A3(1,:)).^2)); 5 S4(i) = sqrt(sum((Y(I,:) - A4(1,:)).^2)); 6 end;</pre>	• Line 3 to 7 $\rightarrow O(I)$ • Line 2 and 8 $\rightarrow O(I)$ • Line 1 $\rightarrow n$. $O(I) = O(n)$ • Line 9 $\rightarrow O(I)$
(a)	(b)

FIGURE 3. Algorithm complexity for stage 3 (a) V distance to A1 and A2 (b) the complexity analysis.

The result of algorithm complexity in stage 3 for V distance to A1, A2, A3 and A4 is $O(n)$.

<pre> 1 V = S2./(S1+S2)</pre>	• Line-1 $\rightarrow O(I)$
(a)	(b)

FIGURE 4. Algorithm complexity for stage 4 (a) ranking and (b) the complexity analysis.

Then, the result of algorithm complexity in stage 4 for ranking is $O(I)$.

ANALYSIS OF GROUP DECISION-MAKING

Data Complexity

There are four (4) decision makings (DM) on this problem in providing alternative assessment for ranking on criteria. Each DM has 4 (four) parameters and alternatives to make the ranking. DM output has 4 (four) alternatives in making decisions on this problem. Meanwhile, every DM has the same assessment criteria and different weights. The weight (W) of DM1 is $W = (2, 3, 5, 4)$, DM2 $W = (5, 3, 4, 2)$ and DM3 $W = (4, 2, 3, 5)$.

Alternative Suitability Rating of Each First Decision Making Criteria (DM1)

The determination of weight and the alternative result of DM1 use data in the criteria value such as in Table 2.

TABLE 2. Alternative Suitability Rating of Each Parameter

Alternatives	Parameters			
	P1	P2	P3	P4
A1	4	4	5	3
A2	3	3	4	2
A3	5	4	2	2
A4	1	2	3	1

Normalized matrix R_I to DM1 uses equation (1), such as in the following completion:

$$R_{1(1)} = \frac{4}{\sqrt{4^2+3^2+5^2+1^2}} = \frac{4}{\sqrt{51}} = \frac{4}{7,1414} = 0,5601 \quad R_{1(2)} = \frac{4}{\sqrt{4^2+3^2+4^2+2^2}} = \frac{4}{\sqrt{45}} = \frac{4}{6,7082} = 0,5962$$

$$R_{1(3)} = \frac{5}{\sqrt{5^2+4^2+2^2+3^2}} = \frac{5}{\sqrt{54}} = \frac{5}{7,3484} = 0,6804 \quad R_{1(4)} = \frac{3}{\sqrt{3^2+2^2+2^2+1^2}} = \frac{3}{\sqrt{18}} = \frac{3}{4,2426} = 0,7071$$

The normalized matrix R_2 is:

$$R_{2(1)} = \frac{3}{\sqrt{4^2+3^2+5^2+1^2}} = \frac{3}{\sqrt{51}} = \frac{3}{7.1414} = 0,4200 \quad R_{2(2)} = \frac{3}{\sqrt{4^2+3^2+4^2+2^2}} = \frac{3}{\sqrt{45}} = \frac{3}{6.7082} = 0,4472$$

$$R_{2(3)} = \frac{4}{\sqrt{5^2+4^2+2^2+3^2}} = \frac{4}{\sqrt{54}} = \frac{4}{7.3484} = 0,5443 \quad R_{2(4)} = \frac{2}{\sqrt{3^2+2^2+2^2+1^2}} = \frac{2}{\sqrt{18}} = \frac{2}{4.2426} = 0,4714$$

The normalized matrix R_3 is:

$$R_{3(1)} = \frac{5}{\sqrt{4^2+3^2+5^2+1^2}} = \frac{5}{\sqrt{51}} = \frac{5}{7.1414} = 0,7001 \quad R_{3(2)} = \frac{4}{\sqrt{4^2+3^2+4^2+2^2}} = \frac{4}{\sqrt{45}} = \frac{4}{6.7082} = 0,5962$$

$$R_{3(3)} = \frac{4}{\sqrt{5^2+4^2+2^2+3^2}} = \frac{4}{\sqrt{54}} = \frac{4}{7.3484} = 0,2721 \quad R_{3(4)} = \frac{2}{\sqrt{3^2+2^2+2^2+1^2}} = \frac{2}{\sqrt{18}} = \frac{2}{4.2426} = 0,4714$$

While the normalized matrix R_4 is as follow:

$$R_{4(1)} = \frac{1}{\sqrt{4^2+3^2+5^2+1^2}} = \frac{1}{\sqrt{51}} = \frac{1}{7.1414} = 0,1400 \quad R_{4(2)} = \frac{2}{\sqrt{4^2+3^2+4^2+2^2}} = \frac{2}{\sqrt{45}} = \frac{2}{6.7082} = 0,2981$$

$$R_{4(3)} = \frac{3}{\sqrt{5^2+4^2+2^2+3^2}} = \frac{3}{\sqrt{54}} = \frac{3}{7.3484} = 0,4082 \quad R_{4(4)} = \frac{1}{\sqrt{3^2+2^2+2^2+1^2}} = \frac{1}{\sqrt{18}} = \frac{1}{4.2426} = 0,2357$$

$$R = \begin{bmatrix} 0,5601 & 0,5962 & 0,6804 & 0,7071 \\ 0,4200 & 0,4472 & 0,5443 & 0,4714 \\ 0,7001 & 0,5962 & 0,2721 & 0,4714 \\ 0,1400 & 0,2981 & 0,4082 & 0,2357 \end{bmatrix} \quad W = (2, 3, 5, 4) \quad Y = \begin{bmatrix} 1,1202 & 1,7888 & 3,4020 & 2,8284 \\ 0,8401 & 1,3416 & 2,7216 & 1,8856 \\ 1,4002 & 1,7888 & 1,3608 & 1,8856 \\ 0,2800 & 0,8944 & 2,0412 & 0,9428 \end{bmatrix}$$

To determine (PIS) A^+ value and (NIS) A^- value, the normalized rating weight (W) is determined using equation (2) to (6).

$$\begin{aligned} y_1^+ &= \max\{1,1202; 0,8401; 1,4002; 0,2800\} = 1,4002 & y_1^- &= \min\{1,1202; 0,8401; 1,4002; 0,2800\} = 0,2800 \\ y_2^+ &= \max\{1,7888; 1,3416; 1,7888; 0,8944\} = 1,7888 & y_2^- &= \min\{1,7888; 1,3416; 1,7888; 0,8944\} = 0,8944 \\ y_3^+ &= \max\{3,4020; 2,7216; 1,3608; 2,0412\} = 3,4020 & y_3^- &= \min\{3,4020; 2,7216; 1,3608; 2,0412\} = 1,3608 \\ y_4^+ &= \max\{2,8284; 1,8856; 1,8856; 0,9428\} = 2,8284 & y_4^- &= \min\{2,8284; 1,8856; 1,8856; 0,9428\} = 0,9428 \end{aligned}$$

$$A^+ \text{ (PIS)} = \{1,4002; 1,7888; 3,4020; 2,8284\}$$

$$A^- \text{ (NIS)} = \{0,2800; 0,8944; 1,3608; 0,9428\}$$

While the result for each weight value of alternative PIS use equation (7) S_i^+ and NIS use equation (8) S_i^- value is:

$$\begin{aligned} S_1^+ &= 0,2800 & S_1^- &= 4,3844 \\ S_2^+ &= 1,3658 & S_2^- &= 2,1270 \\ S_3^+ &= 2,2484 & S_3^- &= 0,6783 \\ S_4^+ &= 2,7317 & S_4^- &= 1,2837 \end{aligned}$$

The completion of this problem ranking model uses equation (9). The problems to DM1 produce alternative (V) for each ideal solutions (IS), i.e. first rank is known from the proximity of alternative contained in V1 with interpretability score of **0.9399**. The result of output ranking is as in Table 3.

TABLE 3. Proximity output of each alternative DM1.

Approaches	Interpretability Scores	Rank
V1	0.9399	1
V2	0.6089	2
V3	0.2317	4
V4	0.3197	3

Alternative Suitability Rating of Each Second Decision Making Criteria (DM2) and Each Third Decision Making Criteria (DM3)

The result of DM2 output uses weight (w) = (4, 3, 1, 2). The problem in DM2 results in alternative (V7) in each ideal solution (IS), i.e. first rank with alternative proximity in V3 and with Interpretability score of **0.8290**. It is shown in Table 4. The output from the input problems in DM3 uses weights (w) = (3, 5, 2, 1). Then it generates alternative proximity output contained in V1 with interpretability score of **0.8778**, as listed in Table 5.

TABLE 4. Proximity output of each alternative DM2

Approaches	Interpretability score	Rank
V1	0. 8009	2
V2	0. 4271	3
V3	0. 8290	1
V4	0. 1436	4

TABLE 5. Proximity output of each alternative DM3

Approaches	Interpretability score	Rank
V1	0. 8778	1
V2	0. 5328	3
V3	0. 7729	2
V4	0. 3716	4

Result of Rating to Each Group Decision Making (GDM)

In this section, we performed steps like ranking, weighting each DM to determine the results of DM ranking in the decision-making. Thus, the ideal DM will be determined in this problem decision-making. The input parameter of each DM consists of 4 alternatives and 3 group DM for assessment, i.e. as in Table 6.

TABLE 6. GDM suitability rating in each alternative

DM Groups	Alternatives			
	A1	A2	A3	A4
DM1	1	2	4	3
DM2	2	3	1	4
DM3	1	3	2	4

Input data of each DM consists of 4 alternatives using weights (W): DM1= 0.5; DM2= 0.3; DM3= 0.2. This DM ranking also uses TOPSIS algorithm in equation (9), and DM ranking result is as in Table 7.

TABLE 7. Proximity output of each alternative GDM

GDM Approaches	DM Interpretability Score	Rank
DM1	0. 4979	3
DM2	0. 6108	1
DM3	0. 5779	2

CONCLUSION

Group decision-making can produce precision and efficiency in making the decision jointly based on the size of O -notation. The result of searching using DM1 is the best ranking in V1 with proximity rating of **0.9399**. DM2 has proximity rating of 0.8290, and DM3 has V1 with proximity rating of 0.8778. Whereas the result of output proximity of each alternative GDM in DM2 with proximity rating to score with interpretability score of **0, 6108**. The Complexity Analysis uses Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) algorithm. The complexity value resulted from the calculation of attribute normalization and weighting is $O(n^2)$. The complexity of positive-negative ideal solution (PIS-NIS) and V distance to A1, A2, A3, A4 is $O(n)$. The algorithm complexity ranking results in $O(1)$. The result of this TOPSIS algorithm can be applied to help to solve problems in

group decision-making, such as the implementation of GDM for determining the location of estates involving various stakeholders or GDM that involves the election of the rector of the faculty senate and the university.

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